

No. of Functional Features Included in the Solution

Date	07/07/2026
Team ID	SWTID-2026-5714
Project Name	OptiCrop: Smart Agriculture Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Crop Prediction, User Input Form, ML Model Loading, Data Validation, Crop Recommendation Display
Additional / Nice-to-Have Features	GitHub Repository, Flask Multi-Page Interface (Home / About / Predict Crop / Result)
Features Tested & Verified	8 / 8 (100%)

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	User Input Form	Accepts Nitrogen, Phosphorus, Potassium, temperature, humidity, pH, and rainfall	Flask HTML Form (predict.html)	Done	High
2	Data Validation	Validates and type-casts user input before prediction	Flask Backend (app.py)	Done	Medium
3	Crop Prediction	Predicts the most suitable crop using the trained ML model	Prediction Engine (Random Forest)	Done	High
4	ML Model Loading	Loads the trained model and scaler (.pkl files) at startup	Model Loader	Done	High
5	Crop Recommendation Display	Displays the predicted crop to the user	Output Module (result.html)	Done	Medium

6	Dataset Integration	Uses the agricultural dataset for training and prediction	Dataset Loader	Done	Medium
7	GitHub Repository	Source code managed using GitHub	Version Control	Done	Low
8	Flask Deployment	Runs the application using Flask, deployed on Render	Application Interface	Done	Medium

Feature Summary:



Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home page, Predict Crop page, Prediction Result page
2	Backend / Logic	3	Crop prediction model, Input validation, Model loading
3	Database / Storage	1	Crop dataset (CSV), Trained model (.pkl)
4	API / Integration	1	Flask routes integrating with the ML model
5	Security / Authentication	0	Not implemented